

2026-06-10-impl-batch2-review

Wave-2 Batch — Independent Code-Review Report (§11.4.125 / §11.4.142)

- **Date:** 2026-06-10
 - **Reviewer:** Independent code-review subagent (READ-ONLY on code; no edits/commits/push)
 - **Repo root:** /Volumes/T7/Projects/HelixCode
 - **Scope:** UNCOMMITTED wave-2 batch — SP4 (helix_agent real os/exec), SP1-cont (helix_code/internal/llm dynamic GetModels + keyrecognition + honest-empty), F1 (git hooks), docs-SQL (architecture docs).
-

VERDICT: GO-WITH-FIXES

The batch is functionally sound, anti-bluff-genuine, and free of regressions. All required verifications PASS. Two non-blocking correctness findings and one scope-hygiene finding warrant attention but do NOT block the commit. There are **zero BLOCKERS**.

- **Finding counts:** 0 BLOCKER · 2 SHOULD-FIX (non-blocking) · 2 NICE-TO-HAVE · 1 SCOPE-NOTE
-

1. Builds + vet — PASS

```
helix_agent: go build ./internal/clis/... BUILD_EXIT=0
helix_agent: go vet   ./internal/clis/...  VET_EXIT=0
helix_code  : go build ./internal/llm/...  BUILD_EXIT=0
helix_code  : go vet   ./internal/llm/...  VET_EXIT=0
```

2. RED→GREEN real — PASS

SP4 standing pin guards (default GREEN polarity):

```
--- PASS: TestD6_QwenCode_GenerateExecsRealBinary (0.45s)    [fake-binary m
--- PASS: TestD6_QwenCode_AbsentBinaryIsHonestError (0.00s)
--- PASS: TestD9_InstanceManager_ExecuteExecsRealBinary (aider/claude_code
--- PASS: TestD9_InstanceManager_AbsentBinaryIsHonestError (all 5 PASS)
```

SP4 under PIN_STUB_BLUFF=1 (§11.4.115 RED reproduction; honest-error path on fixed artifact) — **PASS**:

```
--- PASS: TestD6_QwenCode_GenerateIsStubBluff (1.08s)
--- PASS: TestD9_InstanceManager_ExecuteStubsAreBluffs (9.48s)    [claude_c
```

SP1-cont new tests (RED_MODE=0 default GREEN):

```

--- PASS: TestKeyRecognition_MultiAliasTable_T112
--- PASS: TestKeyRecognition_PresentProviders_AliasMatch_T112
--- PASS: TestKeyRecognition_PresentProviders_AbsenceAndPlaceholder_T112
--- PASS: TestKeyRecognition_IsPlaceholder_T112
--- PASS: TestFetchExternalModels_NoVerifier_HonestEmpty_D5
--- PASS: TestOpenAI_DynamicModels_D5 / DeepSeek / Mistral / Anthropic _Dy

```

F1 hook suite:

RESULT: 27 PASS / 0 FAIL (includes the embedded paired §1.1 mutation tes

3. Anti-bluff soundness — PASS (guards are genuine, not tautologies)

- **SP4 §1.1 live mutation (performed + reverted):** reverted `qwencode.go generate()` to its templated literal → `TestD6_QwenCode_GenerateExecsRealBinary` **FAILED** with “marker absent ... BLUFF-001 reintroduced”. Source restored + rebuilt clean. Guard is real.
- **SP1-cont polarity (verified):** `RED_MODE=1` on the `FIXED` tree **FAILS** `TestFetchExternalModels_NoVerifier_HonestEmpty_D5` + all four `*_DynamicModels_D5` (the fix is present, so the RED-reproduction assertion correctly inverts) — proving the polarity switch is genuinely wired, not a constant-true test.
- **F1 mutation:** suite’s own paired mutation (“broken sibling-assertion no longer blocks” → restored “blocks again”) **PASSED** inside the 27/0 run.
- **No INTRODUCED bluffs:** scan of + lines across all production source (`helix_code llm + helix_agent clis, new + modified`, excluding `_test.go`) for `simulated|for now|TODO implement|in production this would` → **clean**. The only + match anywhere is a doc-comment in `instance_manager.go:913` that QUOTES the removed bluff (“<Agent> execution completed”), which the constitutional anti-bluff smoke-check explicitly excludes.
- **Pre-existing** `instance_manager.go:389` “**For now**” confirmed NOT newly introduced (it appears only as unchanged diff context; the diff touches lines ~906–1030 only).

4. No regression — PASS

- Full `dev.helix.code/internal/llm` suite: **ok (63.07s)**, 0 fail.
- Full `dev.helix.agent/internal/clis/...` suite: **0 FAIL** across all packages (`qwencode 12.2s` incl. new guards; `aider/claude/codex/cline/kiro/openhands` all ok).
- **no-network-on-construction contract preserved:** `NewOpenAIProvider / NewDeepSeekProvider / NewMistralProvider` call only `initializeModels()` (seed, no network); `NewAnthropicProvider` seeds via `getAnthropicModels()`. The live `/models` fetch is deferred to the first `GetModels()` call via `sync.Once(refreshCatalogOnce)`. No constructor calls `refreshCatalogOnce / fetchModelCatalog`.
- **model_discovery honest-empty reconciled, not deleted:** the removed `verifier` import is no longer referenced in `model_discovery.go`; `ConvertVerifiedToModelInfo` remains used by `verifier_integration.go:32` (not orphaned); the sibling behaviour is covered by the `NEW model_discovery_fallback_test.go` (reconciliation test), so no test was silently deleted (§11.4.120 satisfied).

- **F1 hooks NOT installed into .git/hooks:** .git/hooks/{pre-commit,commit-msg,post-commit} all ABSENT; submodules/helix_agent/.git/hooks clean. install_git_hooks.sh is operator-gated (“NOT run automatically”); test_hooks.sh exercises hooks in isolated temp dirs only.

5. Scope + CONST-042 — PASS

- **No go.mod / go.sum edits** in root, helix_code, or helix_agent.
- **Lane discipline:** SP4 → helix_agent only; SP1-cont → helix_code/internal/llm only; F1 → scripts/git_hooks + installer; docs-SQL → docs/architecture. No cross-lane bleed.
- **CONST-042 no-secret-leak:** keyrecognition.go does NO logging of any kind (presence-only; reads env, never prints a value). Provider refresh log.Printf lines log only model COUNTS — none reference key/token/secret/apiKey. In openai_compatible_catalog.go the apiKey flows ONLY into the Authorization: Bearer header, never logged. No secret literals in the new docs.
- **docs-SQL:** schema self-declares “Target engine: PostgreSQL 15+ ... DESIGN — NOT a deployed migration” with a §11.4.44 revision header. sqlite3 parse “errors” are expected Postgres-dialect mismatch (TIMESTAMPZ, COMMENT ON TABLE, now()), not real defects.

Findings

SHOULD-FIX (non-blocking)

1. keyrecognition.go:90 — **over-broad placeholder token "xxx"**. placeholderTokens contains the bare substring "xxx", matched case-insensitively via strings.Contains. Any REAL provider key that happens to contain xxx as a substring would be wrongly classified as a placeholder and the provider reported absent (false-negative key presence). sk-xxx in the same list is already subsumed by "xxx" (redundant). Errs toward honest-empty (safe direction, not a false-positive), hence non-blocking — but tighten to an anchored/standalone match (e.g. exact "xxx"/"sk-xxx" token or a regex) so legitimate keys are not dropped. File:line — helix_code/internal/llm/keyrecognition.go:90.
2. provider_dynamic_models_test.go — **does not assert the construction contract**. The four *_DynamicModels_D5 tests build providers via struct literal + direct initializeModels(), never via New*Provider, so they prove the live-fetch path but do NOT regression-guard “no network at construction.” The contract is correct by design (verified manually in §4), but a future refactor that moved the fetch into the constructor would not be caught by these tests. Add one assertion that New*Provider performs no /models request (e.g. an httptest server with a request counter that must read 0 immediately after construction). File — helix_code/internal/llm/provider_dynamic_models_test.go.

NICE-TO-HAVE

1. **Partial executor conversion in instance_manager.go (pre-existing, in-scope-adjacent)**. This batch converts 5 executors to real exec (Aider, ClaudeCode, Codex, Cline, OpenHands) plus Qwen via qwencode.go. The remaining ~34 execute* methods (executeKiro ... executeLlamaCode, lines ~1033–1270+) STILL return hardcoded "<Agent> execution completed" templates — pre-existing stub bluffs, NOT

newly introduced by this batch (the diff only removes 5 such templates). Out of the stated D-6/D-9 scope, but these remain live BLUFF-003-class stubs and should be tracked for a follow-up wave so the conversion is not left half-applied.

- 2. `openai_compatible_catalog.go` **default token limits are heuristic seeds.** `defaultContext/defaultMaxTokens` (e.g. OpenAI 128000/4096) are applied uniformly to every live-fetched model id. Acceptable as a metadata seed (the model NAMES are live per CONST-036), but per-model context/limits remain approximate until the verifier funnel enriches them. Documented honestly in-comment; no action required beyond awareness.

SCOPE-NOTE

- 1. **Untracked** `docs/caf/ (Status.md, map.tsv)` is **NOT part of the 4 stated streams.** It is a repo-fork PLAN log (`gh repo fork ... --org vasic-digital ...`) from unrelated work present in the working tree. Harmless and untracked, but it must NOT be swept into this batch's commit (§11.4.84 quiescence / §11.4.30 staging discipline) — stage the batch files explicitly, never `git add -A`.

Captured build/test PASS-FAIL summary

BUILD	helix_agent	go build ./internal/clis/...	PASS (exit 0)
VET	helix_agent	go vet ./internal/clis/...	PASS (exit 0)
BUILD	helix_code	go build ./internal/llm/...	PASS (exit 0)
VET	helix_code	go vet ./internal/llm/...	PASS (exit 0)
TEST	SP4	pin guards (default GREEN)	PASS (D6 + D9, all
TEST	SP4	pin guards PIN_STUB_BLUFF=1	PASS (D6 + D9)
TEST	SP4	§1.1 live mutation (revert generate→stub)	FAIL-as-required →
TEST	SP1-cont	new tests (RED_MODE=0)	PASS (KeyRecognitio
TEST	SP1-cont	polarity RED_MODE=1 (fixed tree)	FAIL-as-required (×
TEST	F1	scripts/git_hooks/test_hooks.sh	PASS (27 PASS / 0 F
TEST	FULL	internal/llm regression	PASS (ok, 63.07s, 0
TEST	FULL	internal/clis regression	PASS (0 FAIL across
CHECK	.git/hooks	NOT polluted	PASS (pre-commit/co
CHECK	go.mod/go.sum	untouched	PASS (none)
CHECK	anti-bluff	INTRODUCED-line scan	PASS (clean; only a
CHECK	CONST-042	no-secret-leak	PASS (presence-only

VERDICT: GO-WITH-FIXES — 0 BLOCKER, 2 SHOULD-FIX, 2 NICE-TO-HAVE, 1 SCOPE-NOTE.